Measles Susceptibility in Glasgow Primary School Children

DAS Group 15

2026-03-17

1 Introduction

Since the late 1980s, the measles, mumps, and rubella (MMR) vaccine has been the foundation of childhood vaccination in the UK. However, a now-discredited report by Andrew Wakefield in 1998 that proposed a connection between the vaccine and autism badly damaged public trust. Due to this false information, vaccination rates dropped to almost 80% in some areas by 2003, which subsequently caused serious measles epidemics.

Even if the Wakefield report was partially and fully retracted in 2004 and 2010, respectively, the timeframe for the restoration of public confidence and the ensuing decline in measles susceptibility is still crucial for analysis. This study assesses the measles susceptibility of primary school students in 133 intermediate zones (IZs) in Glasgow between 1998 and 2012 using data from the Scottish Childhood Immunisation Record System (SCIRS).

This study evaluates the probabilities of susceptibility in each post-1998 year directly against the 1998 baseline, avoiding the assumption of a linear trend by treating each year as a categorical component. In particular, we want to address:

1.Did Glasgow exhibit a statistically significant change in measles susceptibility relative to the 1998 baseline?

2.Did the turning point in susceptibility coincide with the 2004 partial retraction, or was there a measurable lag in public health recovery

2 Exploratory Data Analysis

Table 1: Summary statistics for the proportion of primary school children susceptible to measles, by year.

		Proportion susceptible			
Year	IZs	Mean	SD	Min	Max
1998	133	0.0488	0.0322	0.0104	0.2581
2000	133	0.0599	0.0466	0.0000	0.3676
2002	133	0.0725	0.0435	0.0137	0.2734
2004	133	0.0680	0.0331	0.0000	0.2203
2008	133	0.0811	0.0504	0.0000	0.2787
2010	133	0.0518	0.0334	0.0000	0.1714
2012	133	0.0474	0.0328	0.0000	0.1923

Table 2: Summary statistics for the proportion of primary school children susceptible to measles, by year.

	Percenatge susceptible (%)			
Year	Mean	SD	Min	Max
1998	4.88	3.22	1.04	25.81
2000	5.99	4.66	0.00	36.76
2002	7.25	4.35	1.37	27.34
2004	6.80	3.31	0.00	22.03
2008	8.11	5.04	0.00	27.87
2010	5.18	3.34	0.00	17.14
2012	4.74	3.28	0.00	19.23

Table 1 shows how the mean proportion of susceptible children changed over time. There is an increase from 1998 through to around 2003, followed by a decline from 2004 onwards, which lines up with the timing of the Wakefield retraction.

Figure 1 shows the overall pattern of susceptibility rising from 1998, peaks around 2003, then falls after 2004. This is consistent with the drop in vaccination rates following the 1998 paper, and a recovery in uptake after the retraction. However, the change in the overall trend does not occur in 2004 directly. There appears to be an overall increase in susceptibility until 2008. That is, there appears to be a delayed effect.

Figure 2 shows the distribution across IZs for each year. There is moderate variation between IZs, which makes sense given that different areas will have different vaccination uptake.

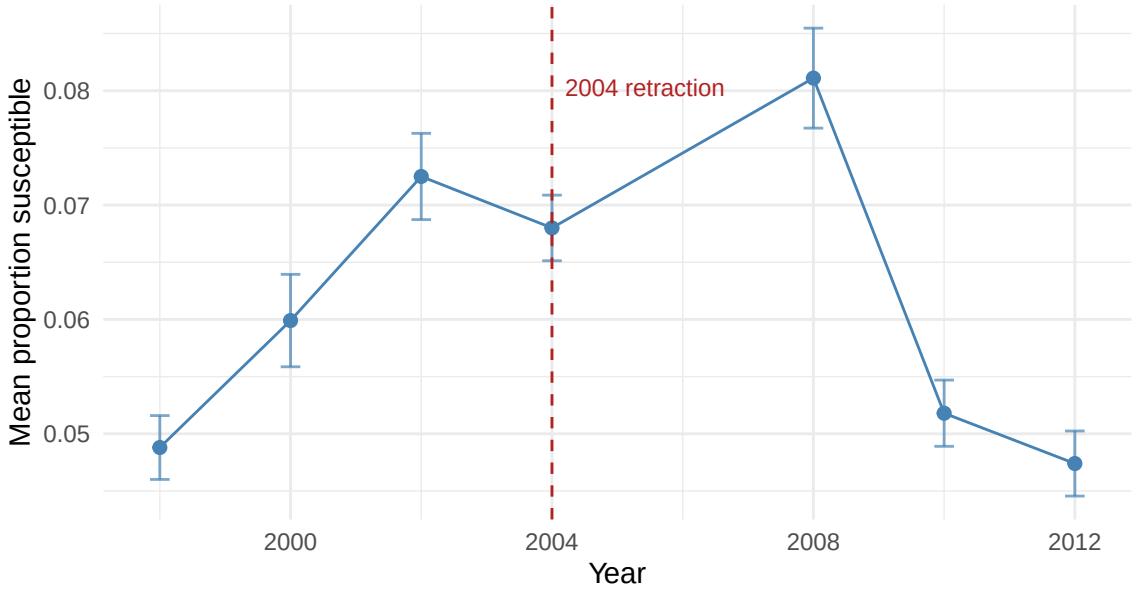


Figure 1: Mean proportion susceptible per year across all 133 Glasgow IZs. Error bars show +/- 1 standard error. The dashed line marks 2004.

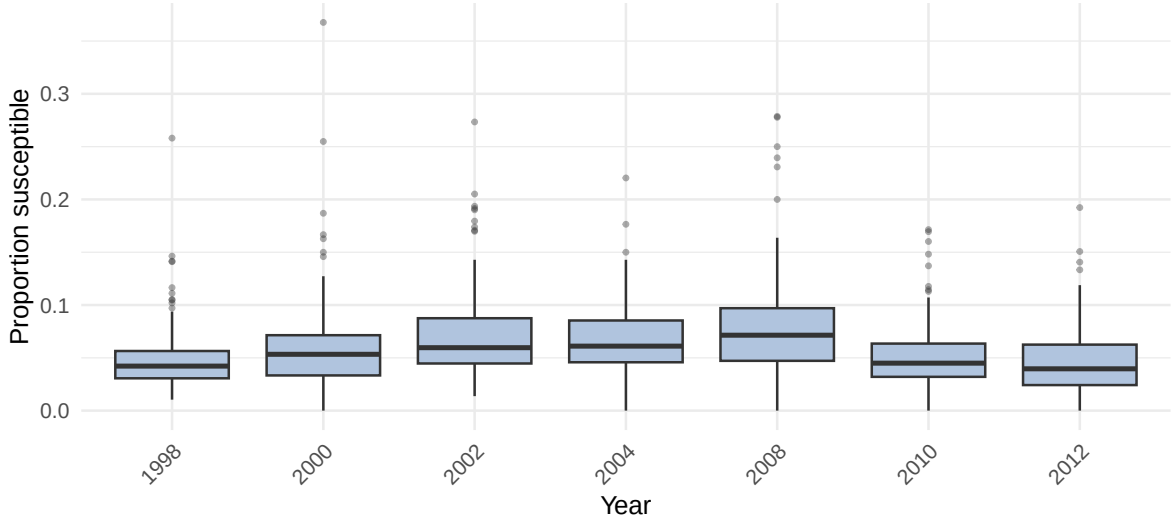


Figure 2: IZ-level proportion susceptible by year. Each box summarises the 133 IZs for that year.

3 Formal Analysis

3.1 Model

Since our outcome is a count of susceptible children out of a total (i.e., Y out of N), the model proposed is a binomial GLM with a logit link, which is a logistic regression model but for proportional count data rather than individual 0/1 outcomes.

We want to test whether the trend in susceptibility changed at 2004, so we fit a model with an interaction term between year and the post-2004 indicator:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \cdot \text{Year}_i^* + \beta_2 \cdot \mathbb{1}[\text{post}] + \beta_3 \cdot (\text{Year}_i^* \times \mathbb{1}[\text{post}])$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \cdot \mathbf{1}[2000] + \beta_2 \cdot \mathbf{1}[2002] + \beta_3 \cdot \mathbf{1}[2004] + \beta_4 \cdot \mathbf{1}[2008] + \beta_5 \cdot \mathbf{1}[2010] + \beta_6 \cdot \mathbf{1}[2012]$$

where Year^* is the year centred at 2004, and $\mathbb{1}[\text{post}]$ equals 1 for 2004 and later. The interaction β_3 captures whether the slope differs before vs after 2004. Here, π_i is the probability of a child being susceptible to measles in year level i.

Fitting the this proposed model produced the following summary output.

Table 3: Table 2: Model summary for binomial model centred at 2004.

	cbind(Y,N-Y)	
Predictors	Odds Ratios	p
(Intercept)	0.10	**<0.001**
yr c	1.12	**<0.001**
post	0.81	**0.001**
yr c × post	0.85	**<0.001**
Observations	931	
Deviance	2044.480	
AIC	4562.514	

```
Dispersion ratio: 2.205


# Overdispersion test

 dispersion ratio =   1.849
          p-value = < 0.001
```

The residual deviance of the binomial model (2044.5 on 927 df) gives a dispersion ratio of approximately 2.2, which exceeds the expected value of 1 under a correctly specified binomial model. This suggests the presence of overdispersion, meaning the standard errors from the binomial model may be too narrow.

Table 4: Table 3: Model summary for binomial model centred at 2008.

	cbind(Y,N-Y)	
Predictors	Odds Ratios	p
(Intercept)	0.10	**<0.001**
yr c	1.06	**<0.001**
post	0.81	**0.002**
yr c × post	0.81	**<0.001**
Observations	931	
Deviance	2002.928	
AIC	4529.192	

3.2 Modelling Measles Susceptibility by Year

3.2.1 Binomial Logistic Regression Model

To assess variation in measles susceptibility across years, a binomial logistic regression model was fitted:

$$Y_i \sim \text{Binomial}(N_i, p_i)$$

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \cdot \text{Year}_{2000,i} + \beta_2 \cdot \text{Year}_{2002,i} + \cdots + \beta_6 \cdot \text{Year}_{2012,i}$$

where (p_i) is the probability that a child in area (i) is susceptible to measles, and 1998 is treated as the reference category.

$$\hat{\phi} = \frac{\text{Residual Deviance}}{\text{Residual Degrees of Freedom}} = 2.128$$

Overdispersion was assessed using the binomial model, with the ratio of residual deviance to degrees of freedom exceeding 1, indicating the presence of over dispersion. To account for this, a quasibinomial model was fitted, which adjusts the standard errors to provide more reliable inference.

To account for overdispersion, a quasibinomial model was fitted:

$$Y_i \sim \text{Quasi-Binomial}(N_i, p_i, \phi)$$

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \cdot \text{Year}_{2000,i} + \cdots + \beta_6 \cdot \text{Year}_{2012,i}$$

Table 5: Odds Ratios for Year-on-Year Susceptibility Relative to 1998 Baseline

	cbind(Y,N-Y)	
Predictors	Odds Ratios	p
(Intercept)	0.05	**<0.001**
Year [2000]	1.26	**<0.001**
Year [2002]	1.57	**<0.001**
Year [2004]	1.42	**<0.001**
Year [2008]	1.70	**<0.001**
Year [2010]	1.09	0.118
Year [2012]	0.98	0.694
Observations	931	
Deviance	1966.583	
AIC	4508.119	

The year-specific logistic regression model shows that measles susceptibility varied significantly across years in Glasgow. Compared with the baseline year (1998), susceptibility increased significantly in 2000, 2002, 2004 and 2008, with the largest increase occurring in 2008 (OR = 1.70, $p < 0.001$). However, there was no significant difference in 2010 or 2012. This suggests that Glasgow did experience changes in measles susceptibility over time.

However, the results do not indicate a clear change occurring specifically in 2004, the year of the Wakefield article retraction. Instead, susceptibility appears to continue increasing after 2004 and peaks around 2008, suggesting the effect was not immediate.

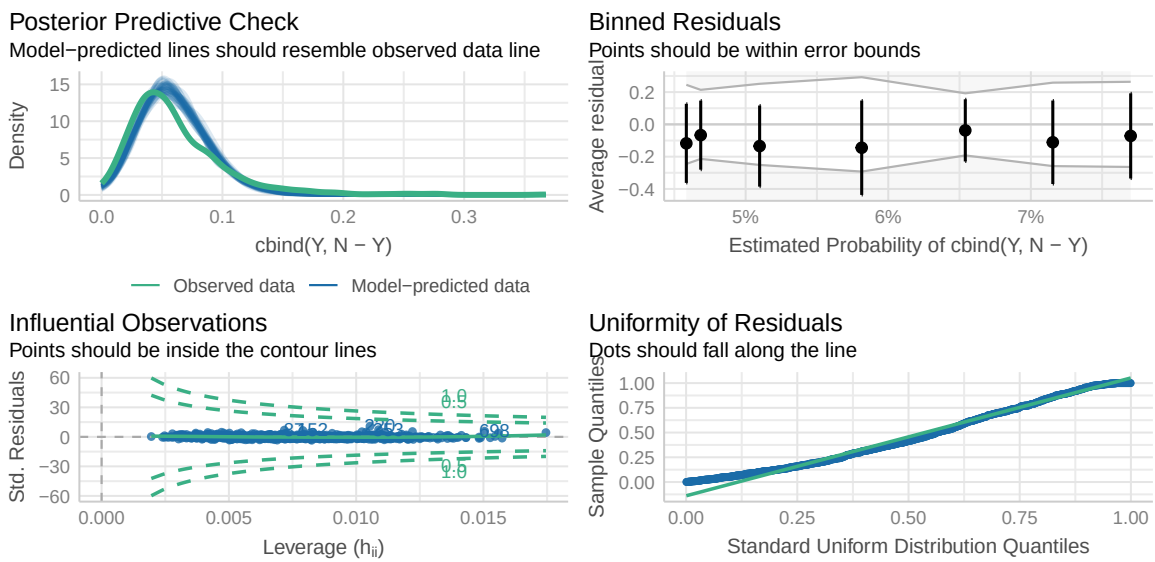
To determine if "Year" is a significant predictor of susceptibility, we compared our model against a null model using a likelihood ratio test:

$$\chi^2 = D_{\text{null}} - D_{\text{model}} = 2179.6 - 1966.6 = 213.0$$

$$df = 930 - 924 = 6$$

$$p = P\left(\chi^2_6 \geq 213.0\right) \approx 3.23 \times 10^{-43}$$

The reduction in deviance from 2179.6 to 1966.6 corresponds to $\chi^2 = 213.0$ with 6 degrees of freedom ($p \approx 3.23 \times 10^{-43}$), providing strong evidence that including year significantly improves model fit, confirming that Glasgow exhibited a clear change in measles susceptibility between 1998 and 2012. This result is identical for both the binomial and quasibinomial models, as the quasibinomial approach adjusts only the standard errors and not the fitted values or deviance.



The diagnostic plots indicate that the year-based binomial model provides an adequate fit to the data. The posterior predictive check shows good agreement between observed and fitted distributions, while binned residuals exhibit no systematic deviations. There is no evidence of influential observations, and residuals appear well-calibrated. Minor discrepancies in the upper tail suggest slight underestimation of extreme values, but these do not materially affect the overall model fit. These plots are the same for the year-based quasibinomial model.

4 Consclusion